Abhijeet Anand

AbhijeetAnand@lbl.gov, https://abhi0395.github.io

Postdoctoral Scientist, Lawrence Berkeley National Lab, Berkeley, CA

Positions

Sep'22 - Aug'25: Postdoctoral Scientist, Lawerence Berkeley National Lab (LBNL), Berkeley, CA, USA

Aug'22: Bridge Postdoctoral Fellow, Max Planck Institute for Astrophysics, Garching, Germany

Sep'18 - Jul'22: IMPRS PhD Fellow, Max Planck Institute for Astrophysics, Garching, Germany

Education

Sep 2018 - Jul 2022: PhD in Astrophysics, Ludwig Maximilian University - Max Planck Institute for Astrophysics (MPA), Germany.

- Thesis: Probing cool and warm circumgalactic gas in galaxies and clusters with large spectroscopic and imaging surveys
- Advisors: Prof. Dr. Guinevere Kauffmann & Dr. Dylan Nelson

Jul 2016 - Jul 2017: MS, Physics, Indian Institute of Science (IISc), Bangalore, India.

- Thesis: A sensitive search for HI 21 cm emission from super disks in radio galaxies
- Advisor: Prof. Nirupam Roy

Aug 2012 - June 2016: BSc (Research), Physics, Indian Institute of Science, Bangalore, India.

- Thesis: Sources of Continuum Opacity in Hydrogen deficient stars
- Advisors: Prof. Gajendra Pandey, Indian Institute of Astrophysics (IIA)

Publications

[40] Napolitano, Lucas; Myers, Adam D.; Tedeschi, Adam; **Anand, Abhijeet**; Herrera-Alcantar, Hiram K.; *et al.*, 2026, *The Composite Spectrum of QSO Absorption Line Systems in DESI DR2*, The Astronomical Journal, **172**, 250 (arXiv:2512.02992) [**Impact Factor: 5.8; Scopus Quartile: Q1**]

[39] Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Siudek, Malgorzata; Aguilar, J.; *et al.* (incl. **A. Anand**), 2026, *Tracing the Cosmic Evolution of the Cool Circumgalactic Medium of Luminous Red Galaxies with DESI Year 1 Data*, The Astrophysical Journal, **1009**, 121 (arXiv:2512.03845) [4 citations] [**Impact Factor: 5.7; Scopus Quartile: Q1**]

[38] Lange, J. U.; Wells, A.; Hearin, A.; Beltz-Mohrmann, G.; Leauthaud, A.; *et al.* (incl. **A. Anand**), 2026, *Cosmological Constraints from Full-scale Clustering and Galaxy-Galaxy Lensing with DESI DR1*, The Astrophysical Journal, **1010**, 170 (arXiv:2512.15962) [**Impact Factor: 5.7; Scopus Quartile: Q1**]

[37] Uzsoy, Ana Sofía M.; Dey, Arjun; Raichoor, Anand; Finkbeiner, Douglas P.; Ramakrishnan, Vandana; *et al.* (incl. **A. Anand**), 2026, *Effect of Local Environment on Lyα Line Profile in DESI/ODIN LAEs*, The Astrophysical Journal, **1010**, 76 (arXiv:2511.17498) [**Impact Factor: 5.7; Scopus Quartile: Q1**]

[36] Blanco, Diana; Leauthaud, Alexie; Lange, Johannes Ulf; Wright, Angus H.; Hildebrandt, Hendrik; *et al.* (incl. **A. Anand**), 2026, *The Power of DESI for Photometric Redshift Calibration:*

A Case Study with KiDS-1000, The Open Journal of Astrophysics, **9**, 63965 (arXiv:2512.15964) [3 citations] [**Impact Factor: 2.4; Scopus Quartile: Q1**]

35 McArthur, E.; Millon, M.; Powell, M.; Wechsler, R. H.; Pan, Z.; *et al.* (incl. **A. Anand**), 2026, *Quasars Acting as Strong Lenses Found in DESI DR1*, The Astrophysical Journal, **1006**, 147 (arXiv:2511.02009) [1 citations] [**Impact Factor: 5.7; Scopus Quartile: Q1**]

34 Crespi, Andrea; Percival, Will J.; Krolewski, Alex; Bonici, Marco; Zhang, Hanyu; *et al.* (incl. **A. Anand**), 2026, *Baryon fraction from the BAO amplitude: A consistent approach to parametrizing perturbation growth*, Physical Review D, **114**, 043545 (arXiv:2511.23459) [1 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

33 Dey, Biprateep; Newman, Jeffrey A.; Zhang, Tianqing; Aguilar, J.; Ahlen, S.; *et al.* (incl. **A. Anand**), 2026, *Deep Spectroscopy with DESI for Photometric Redshift Training and Calibration*, The Astronomical Journal, **172**, 65 (arXiv:2604.06143) [2 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

32 Moore, Samuel G.; Cole, Shaun; Wilson, Michael; Norberg, Peder; Moustakas, John; *et al.* (incl. **A. Anand**), 2026, *DESI DR2 Galaxy luminosity functions*, Monthly Notices of the Royal Astronomical Society, **549**, stag866 (arXiv:2511.01803) [10 citations] [**Impact Factor: 5.2; Scopus Quartile: Q1**]

31 DESI Collaboration; Abdul Karim, M.; Adame, A. G.; Aguado, D.; Aguilar, J.; *et al.* (incl. **A. Anand**), 2026, *Data Release 1 of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **171**, 285 (arXiv:2503.14745) [705 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

30 Turner, Wynne; Cuceu, Andrei; Martini, Paul; Aguilar, J.; Ahlen, S.; *et al.* (incl. **A. Anand**), 2026, *Probing the limits of cosmological information from the Lyman-α forest 2-point correlation functions*, Journal of Cosmology and Astroparticle Physics, **2026**, 010 (arXiv:2509.14322) [4 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

29 Zaborowski, E. A.; Taylor, P.; Honscheid, K.; Cuceu, A.; de Mattia, A.; *et al.* (incl. **A. Anand**), 2026, *H ₀ without the sound horizon (or supernovae): A 2% measurement in DESI DR1*, Journal of Cosmology and Astroparticle Physics, **2026**, 004 (arXiv:2510.19149) [10 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

28 Karaçaylı, Naim Göksel; Ravoux, Corentin; Martini, Paul; Goff, Jean-Marc Le; Armengaud, Eric; *et al.* (incl. **A. Anand**), 2026, *DESI DR1 Lyα 1D power spectrum: Validation of estimators*, Journal of Cosmology and Astroparticle Physics, **2026**, 048 (arXiv:2509.13593) [6 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

27 Ebina, H.; White, M.; Raichoor, A.; Dey, Arjun; Schlegel, D.; *et al.* (incl. **A. Anand**), 2026, *Clustering analysis of medium-band selected high-redshift galaxies*, Journal of Cosmology and Astroparticle Physics, **2026**, 019 (arXiv:2509.26467) [7 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

26 Hsu, Yuan-Ming; Huang, Xiaosheng; Storfer, Christopher J.; Inchausti, Jose Carlos; Schlegel, David; *et al.* (incl. **A. Anand**), 2026, *A New Way to Discover Strong Gravitational Lenses: Pairwise Spectroscopic Search from DESI DR1*, The Astrophysical Journal Supplement Series, **282**, 41 (arXiv:2509.16033) [5 citations] [**Impact Factor: 7.7; Scopus Quartile: Q1**]

25 Herrera-Alcantar, Hiram K.; Armengaud, Eric; Yèche, Christophe; Gordon, Calum; Casas, Laura; *et al.* (incl. **A. Anand**), 2025, *The Lyman-α forest from LBGs: First 3D correlation measurement with DESI and prospects for cosmology*, Journal of Cosmology and Astroparticle

Physics, **2025**, 053 (arXiv:2507.21852) [8 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

24 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Alam, S.; Allen, L.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints*, Physical Review D, **112**, 083515 (arXiv:2503.14738) [1712 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

23 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Allende Prieto, C.; Alves, O.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest*, Physical Review D, **112**, 083514 (arXiv:2503.14739) [333 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

22 Brodzeller, A.; Wolfson, M.; Santos, D. M.; Ho, M.; Tan, T.; *et al.* (incl. **A. Anand**), 2025, *Construction of the damped Lyα absorber catalog for DESI DR2 Lyα BAO*, Physical Review D, **112**, 083510 (arXiv:2503.14740) [30 citations] [**Impact Factor: 5.3; Scopus Quartile: Q1**]

21 Rashkovetskyi, Michael; Eisenstein, Daniel J.; Aguilar, Jessica Nicole; Ahlen, Steven; **Anand, Abhijeet**; *et al.*, 2025, *Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect*, The Open Journal of Astrophysics, **8**, 46033 (arXiv:2508.20904) [**Impact Factor: 2.4; Scopus Quartile: Q1**]

20 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 008 (arXiv:2411.12021) [211 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

19 **Anand, Abhijeet**; Aguilar, J.; Ahlen, S.; Bianchi, D.; Brodzeller, A.; *et al.*, 2025, *The Cosmic Evolution of C IV Absorbers at 1.4 < z < 4.5: Insights from 100,000 Systems in DESI Quasars*, The Astrophysical Journal, **990**, 151 (arXiv:2504.20299) [9 citations] [**Impact Factor: 5.7; Scopus Quartile: Q1**]

18 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements*, Journal of Cosmology and Astroparticle Physics, **2025**, 028 (arXiv:2411.12022) [410 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

17 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics*, Journal of Cosmology and Astroparticle Physics, **2025**, 017 (arXiv:2411.12020) [156 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

16 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 III: baryon acoustic oscillations from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 012 (arXiv:2404.03000) [592 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

15 Wu, Xuanyi; Cai, Z.; Lan, T.-W.; Zou, S.; **Anand, Abhijeet**; *et al.*, 2025, *Tracing the Evolution of the Cool Gas in CGM and IGM Environments through Mg II Absorption from Redshift z = 0.75 to z = 1.65 Using DESI-Y1 Data*, The Astrophysical Journal, **983**, 186 (arXiv:2407.17809) [16 citations] [**Impact Factor: 5.7; Scopus Quartile: Q1**]

14 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*, Journal of Cosmology and Astroparticle Physics, **2025**, 021 (arXiv:2404.03002) [2020 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

13 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest*, Journal of Cosmology and Astroparticle Physics, **2025**, 124 (arXiv:2404.03001) [417 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

12 Ross, A. J.; Aguilar, J.; Ahlen, S.; Alam, S.; **Anand, Abhijeet**; *et al.*, 2025, *The construction of large-scale structure catalogs for the Dark Energy Spectroscopic Instrument*, Journal of Cosmology and Astroparticle Physics, **2025**, 125 (arXiv:2405.16593) [79 citations] [**Impact Factor: 6.4; Scopus Quartile: Q1**]

11 Scholte, Dirk; Saintonge, Amélie; Moustakas, John; Catinella, Barbara; Zou, Hu; *et al.* (incl. **A. Anand**), 2024, *The atomic gas sequence and mass-metallicity relation from dwarfs to massive galaxies*, Monthly Notices of the Royal Astronomical Society, **535**, 2341 (arXiv:2408.03996) [37 citations] [**Impact Factor: 5.2; Scopus Quartile: Q1**]

10 Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Napolitano, Lucas; **Anand, Abhijeet**; *et al.*, 2024, *Probing the Impact of Radio-mode Feedback on the Properties of the Cool Circumgalactic Medium*, The Astrophysical Journal, **974**, 191 (arXiv:2405.08314) [9 citations] [**Impact Factor: 5.7; Scopus Quartile: Q1**]

9 Galiullin, Ilkham; Rodriguez, Antonio C.; El-Badry, Kareem; Szkody, Paula; **Anand, Abhijeet**; *et al.*, 2024, *Searching for new cataclysmic variables in the Chandra Source Catalog*, Astronomy and Astrophysics, **690**, A374 (arXiv:2408.00078) [9 citations] [**Impact Factor: N/A; Scopus Quartile: N/A**]

8 **Anand, Abhijeet**; Guy, Julien; Bailey, Stephen; Moustakas, John; Aguilar, J.; *et al.*, 2024, *Archetype-based Redshift Estimation for the Dark Energy Spectroscopic Instrument Survey*, The Astronomical Journal, **168**, 124 (arXiv:2405.19288) [55 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

7 DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *The Early Data Release of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **168**, 58 (arXiv:2306.06308) [620 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

6 DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; *et al.* (incl. **A. Anand**), 2024, *Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **167**, 62 (arXiv:2306.06307) [339 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

5 Napolitano, Lucas; Pandey, Agnesh; Myers, Adam D.; Lan, Ting-Wen; **Anand, Abhijeet**; *et al.*, 2023, *Detecting and Characterizing Mg II Absorption in DESI Survey Validation Quasar Spectra*, The Astronomical Journal, **166**, 99 (arXiv:2305.20016) [34 citations] [**Impact Factor: 5.8; Scopus Quartile: Q1**]

4 Ayromlou, Mohammadreza; Kauffmann, Guinevere; **Anand, Abhijeet**; & White, Simon D. M., 2023, *The physical origin of galactic conformity: from theory to observation*, Monthly Notices of the Royal Astronomical Society, **519**, 1913 (arXiv:2207.02218) [33 citations] [**Impact Factor: 5.2; Scopus Quartile: Q1**]

3 **Anand, Abhijeet**; Kauffmann, Guinevere; & Nelson, Dylan, 2022, *Cool circumgalactic gas in galaxy clusters: connecting the DESI legacy imaging survey and SDSS DR16 Mg II absorbers*, Monthly Notices of the Royal Astronomical Society, **513**, 3210 (arXiv:2201.07811) [33 citations] [**Impact Factor: 5.2; Scopus Quartile: Q1**]

2 **Anand, Abhijeet**; Nelson, Dylan; & Kauffmann, Guinevere, 2021, *Characterizing the*

abundance, properties, and kinematics of the cool circumgalactic medium of galaxies in absorption with SDSS DR16, Monthly Notices of the Royal Astronomical Society, **504**, 65 (arXiv:2103.15842) [70 citations] [**Impact Factor: 5.2; Scopus Quartile: Q1**]

1 **Anand, Abhijeet**; Roy, Nirupam; & Gopal-Krishna, 2019, *Search for H I emission from superdisk candidates associated with radio galaxies*, Research in Astronomy and Astrophysics, **19**, 083 (arXiv:1812.06875) [3 citations] [**Impact Factor: 2.6; Scopus Quartile: Q2**]